# Transformixer:Attention-Based Variate Mixing without Positional Bias for Multivariate Forecasting

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## Outline

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**Task:** multivariate long-term forecasting
lookback $X \in \mathbb{R}^{L \times C} \rightarrow$ horizon $\hat{Y} \in \mathbb{R}^{H \times C}$

## Challenge

Models must capture *temporal* dynamics **and** *cross-variate* dependencies.
Especially hard when $C$ is large (e.g., Traffic: $C$=862) and channels have **no natural order**.

Recent mixers (e.g., xLSTM-Mixer) refine a linear temporal forecast with a joint variate mixer.
But recurrent mixing along channels is order-sensitive.

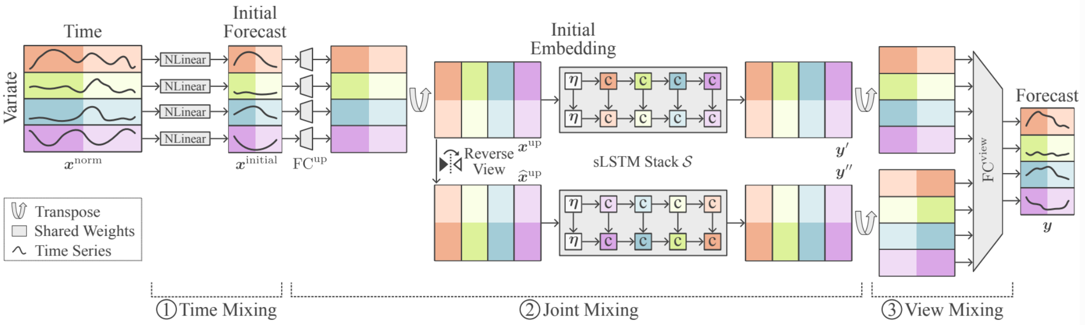
# Background: xLSTM-Mixer

1. RevIN normalization
2. Shared **NLinear** temporal forecast (efficient)
3. **sLSTM** mixes along the *variate* axis (+ memory tokens)
4. Forward + **reverse** multi-view fusion

**Inductive-bias issue**

Channel indices (sensors, meters, roads) are often an **arbitrary ordering**.
Sequential mixing depends on that order and needs a reverse view to reduce artifacts.

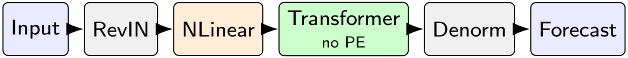
# xLSTM-Mixer architecture




Same temporal pipeline (RevIN + NLinear); Transformixer replaces joint/view mixing with a PE-free Transformer.

# Idea: Transformixer

**Keep** what works: RevIN + shared NLinear temporal mixing.
**Replace** the ordered sLSTM mixer with a PE-free Transformer over variate tokens.




- Variates treated as a **set** (permutation-equivariant)
- Direct pairwise interactions via multi-head attention
- No memory tokens, no forward/reverse multi-view

# vs. xLSTM-Mixer (what changes)

| Component | xLSTM-Mixer | Transformixer |
|---|---|---|
| RevIN + NLinear | yes | yes |
| Variate mixer | sLSTM (+ mem.) | Transformer |
| Positional bias on variates | via recurrence | **none** |
| Forward/reverse views | yes | **no** |
| Memory tokens | yes | **no** |

## Related but different from iTransformer

iTransformer = full inverted backbone.
Transformixer = attention as a *drop-in mixer block* after NLinear.

# Why Transformer mixing? (not only equivariance)

1. **Direct global interactions** — $O(1)$ path length between any pair of variates (sLSTM propagates sequentially along an arbitrary order)
2. **Multi-head specialization** — different dependency patterns in parallel
3. **Simpler architecture** — drop multi-view + memory tokens
4. **Hardware-parallel over $C$** — with honest $O(C^2)$ caveat

Hypothesis: gains should be clearest when $C$ is large (Traffic).

# Experimental setup

**Datasets**

- ECL: $C$=321
- Traffic: $C$=862

**Protocol**

- $L$=$H$=96
- Metrics: MSE, MAE
- Multivariate ($M$)

**Baselines**

- RLinear, Informer
- PatchTST, iTransformer
- xLSTM-Mixer

**Fairness note**

- Baselines: author-recommended hparams
- Transformixer: not tuned (inherits xLSTM-Mixer settings)

**Scope**

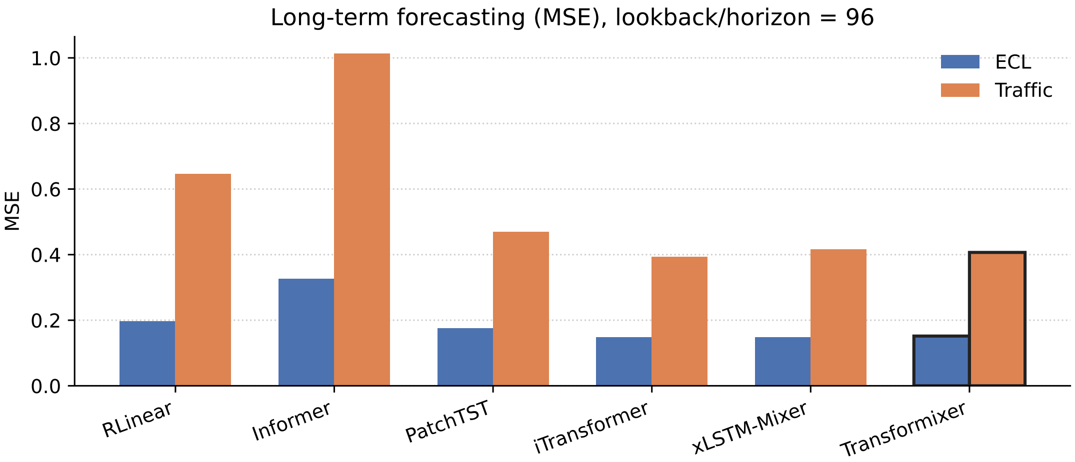
Limited compute/time $\Rightarrow$ 2 datasets, 1 horizon, single runs.
Architectural study, not a SOTA claim.

# Main results

| | ECL | | Traffic | |
|---|---|---|---|---|
| Model | MSE | MAE | MSE | MAE |
| RLinear | 0.1973 | 0.2740 | 0.6463 | 0.3844 |
| Informer | 0.3263 | 0.4068 | 1.0140 | 0.5624 |
| PatchTST | 0.1754 | 0.2603 | 0.4699 | 0.3005 |
| iTransformer | <u>0.1481</u> | 0.2398 | **0.3931** | 0.2687 |
| xLSTM-Mixer | **0.1480** | **0.2352** | 0.4157 | <u>0.2607</u> |
| Transformixer | 0.1517 | <u>0.2364</u> | <u>0.4068</u> | **0.2558** |

- **Traffic:** beats xLSTM-Mixer; **best MAE** overall
- **ECL:** close, slightly behind xLSTM-Mixer / iTransformer ($\sim$2.5% MSE)

# Results at a glance

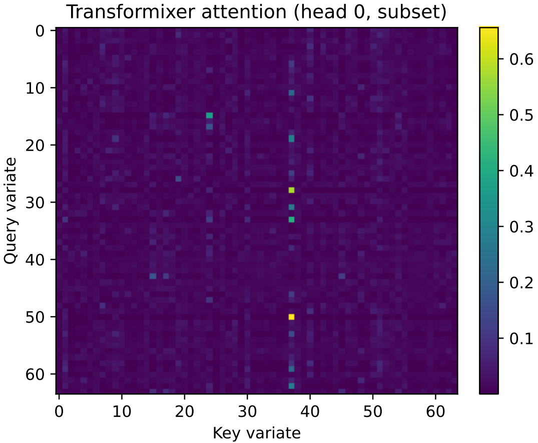


# Permutation probe (Traffic checkpoints)

| | Identity | | Over 9 orders (mean±std) | |
|---|---|---|---|---|
| Model | MSE | MAE | MSE | MAE |
| xLSTM-Mixer | 1.3463 | 0.9224 | 1.3486±0.0015 | 0.9233±0.0005 |
| Transformixer | 1.2841 | 0.9019 | 1.2865±0.0014 | 0.9027±0.0005 |

- Transformixer nearly invariant (equivariance)
- xLSTM mild drift (multi-view helps), but **always worse**
- Relative gap $\approx$ 4.6% MSE / 2.2% MAE on this probe

# Attention visualization




**What we see**

- Sparse, non-uniform attention
- Hub-like key variates
- Not purely diagonal

Supports *direct pairwise* cross-variate mixing.

# Limitations (stated openly)

- Only **2 datasets**, horizon **96**, **single** runs
- Baselines use author-tuned hparams; Transformixer does **not**
- No full multi-horizon / multi-seed study (compute & time)
- Attention cost $O(C^2)$ not profiled

## Still useful

Competitive results + clearer inductive bias for unordered channels.
Strongest relative signal on high-$C$ Traffic.

1. **Transformixer** = NLinear temporal mix + PE-free Transformer variate mixer
2. Order-agnostic *and* direct multi-head cross-variate mixing
3. Competitive with strong baselines; improves on xLSTM-Mixer for Traffic
4. Promising direction under limited compute; broader eval is future work

## Thank you

Questions?